Limits of REDD-Only Evidence for NILM Deployment

Measurement context, preprocessing, and complementary validation

Stephen Makonin Companion technical note, version 1.2
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Purpose and actual-data findings

REDD helped establish open energy-disaggregation research and remains valuable for historical reproduction, benchmark comparisons, and focused algorithmic tests. This update examines the full conventional low-frequency release directly and asks what evidence it supports for deployment across households, seasons, and measurement systems. [1, 14]

All 122 canonical files—116 channel files and six label files—matched the locally archived REDD Version 1.0 release byte for byte. The merged Python 3 importer processed all 56,341,629 source observations successfully. Independent verification confirmed preservation of every one of the 14,113,792 original mains records and checked all 7,755,931 derived timestamp-union rows. No mains value was replaced using target measurements. [14]

The fresh audit found no malformed records, duplicate timestamp keys, or conflicting duplicates. It did find 47,957 descending timestamp adjacencies in source-file order, nonuniform sampling, and long recording gaps. Observed mains modes are one second in houses 1–5 and two seconds in house 6; circuit modes are four seconds in houses 1 and 5 and three seconds elsewhere. Source disorder can be handled by sorting without discarding readings.

Exact all-channel completeness occurs at 1,797,457 of 7,056,896 both-mains timestamps, or 25.47 percent. This is a timestamp-coincidence ratio, not a sensor-loss rate. With a predeclared 60-second gap rule, shared threshold-connected duration ranges from 8.23 to 55.00 percent of each home's mains endpoint envelope. That duration is an availability proxy, not measured continuous uptime or reconstructed ground truth.

These findings do not support a blanket claim of numerical corruption. In particular, no partial constituent group or missing second mains leg was concealed by the historical importer's any-member rule in this archive. These specific partial-group and endpoint failure modes remain conditional implementation risks; the directly observed concerns are temporal coverage, cadence, source order, measurement compatibility, and the scope of deployment claims. The 2,859 negative exact-time numerical residuals were retained unchanged and do not independently establish meter or clock error.

This is a descriptive data-quality audit, not a new NILM accuracy experiment or calibration study. Historical 2015 log results remain separately identified and do not describe the new importer. Stephen Makonin's archived repairs are his software choices, not REDD's acquisition pipeline.

Author disclosure. Stephen Makonin wrote the historical importer examined here and coauthored the AMPds2, RAE, and R1Hz dataset publications. Recommendations identify complementary evaluation roles, without a universal accuracy ranking. Equal measurement, missingness, provenance, and reporting scrutiny applies to every dataset. [7, 10, 13]

1 Fresh raw-archive audit: provenance and integrity

The controlling source is the local low_freq.tar.bz2 archive, 169,367,706 bytes, identified as REDD Version 1.0 by its accompanying README. Its SHA-256 is 9911f3fd7d8c7fe89a3ce4634374917fa52a536103cd203cd6e7fa7ccb0ef420. All 116 channel files and six label files in the canonical house_1 through house_6 directories matched their archive members byte for byte. Local copies, precleaned exports, high-frequency files, and unrelated metadata files were excluded. The checksum identifies this copy; it is not an independently published official checksum. [14]

Measurement-source files total 932,604,931 bytes. The merged importer revision is 4479464, run with Python 3.12.14. All six imports succeeded, with no duplicate deduplication, malformed-row rejection, or conflicting-key failure. Input file hashes were checked against the archive and source files remained unchanged.

Table 1: Actual source inventory and descriptive modal intervals

House	Channels	Source records	Order descents	Mains / circuit mode (s)
1	20	16,549,124	8,169	1 / 4
2	11	5,265,899	997	1 / 3
3	22	10,936,708	15,732	1 / 3
4	20	13,626,212	532	1 / 3
5	26	2,534,252	6,322	1 / 4
6	17	7,429,434	16,205	2 / 3

An order descent means that a record's timestamp is smaller than the preceding record's timestamp in the same source file. The 47,957 descents span 6,528 distinct house/timestamp pairs; neither number is a count of independent clock faults or lost observations. All 116 files still began at their minimum timestamp and ended at their maximum. Thus internal disorder occurred, but the historical first/last endpoint assumption did not fail on this release.

The audit used separate observation masks, unique timestamp keys, exact interval histograms, and disk-backed canonical ordering. An independent verifier checked every original mains record against channels.csv, including its value, source line number, and source order. It also checked every loads.csv row against an independent source index. All 7,056,896 available MAIN values equaled the exact sum of the original two legs, including all 2,859 negative-residual rows. These checks establish numerical preservation; they do not establish meter calibration.

For each channel, source rows, unique unambiguous timestamp keys, and elapsed endpoint span are separate quantities. A calendar-second grid density is not uptime when native cadence varies. The audit protocol was declared before the new aggregate scan, with a primary gap threshold of 60 seconds and sensitivity thresholds of 30, 120, and 300 seconds. Exact intervals and all per-channel results are supplied with the note.

2 Observed temporal coverage and gap sensitivity

For sorted consecutive unique timestamps, let a threshold-connected span include the half-open interval between two observations only when their separation is at most T seconds. Longer intervals are excluded entirely; no duration is added after the last observation. Intersect these interval unions across the required channels to obtain a shared duration proxy. This construction creates no new power readings and does not verify continuous availability between endpoints. [14]

All fractions below use each house's earliest mains start through latest mains end as the denominator. That endpoint envelope combines recording boundaries and internal gaps. A separately supplied common-endpoint denominator isolates the interval where all required channels' endpoints overlap.

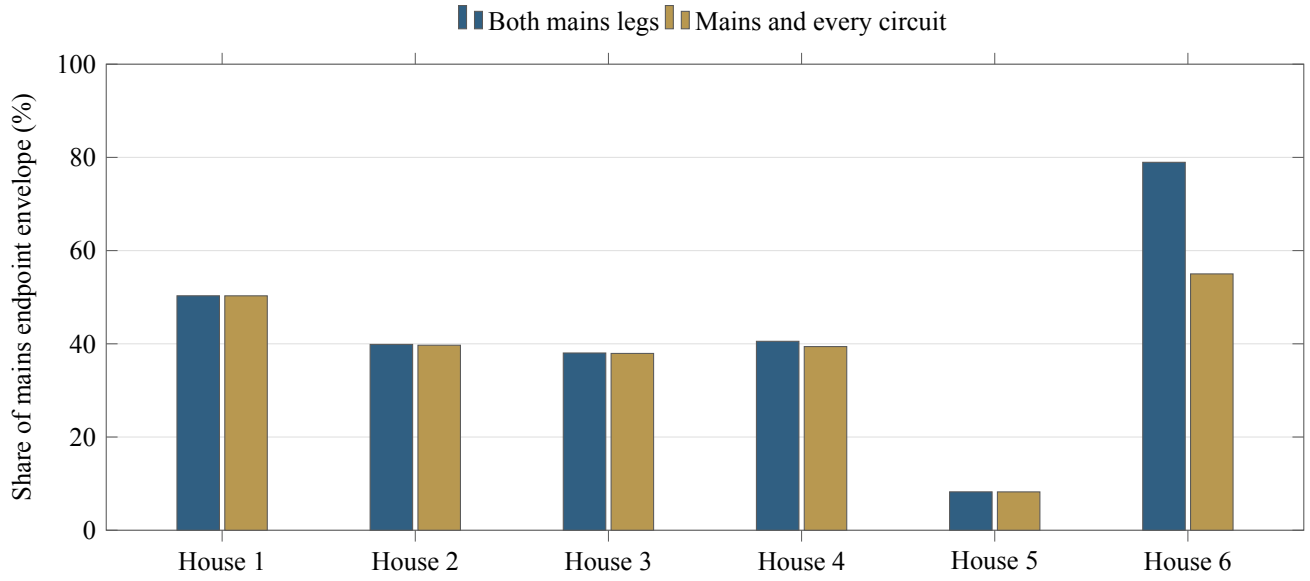


Figure 1: Fresh raw-data audit. Both series use the 60-second rule and the same house-specific mains endpoint envelope. Gold requires every original circuit and both mains legs. These are threshold-connected durations, not measured uptime, energy coverage, or sensor-loss rates.

Table 2: Threshold-connected durations and longest consecutive-record gaps

House	Mains envelope (days)	Both mains, 60 s (days)	All channels, 60 s (days)	All / envelope (%)	Longest mains gap (days)
1	36.27	18.25	18.25	50.30	9.45
2	35.03	13.97	13.91	39.70	20.70
3	44.80	17.04	17.00	37.94	19.07
4	47.98	19.45	18.91	39.41	20.14
5	43.83	3.61	3.61	8.23	31.42
6	23.41	18.48	12.88	55.00	3.94

Pooled as household exposure, the six mains envelopes total 19,986,137 seconds (231.32 house-days); shared all-channel threshold-connected spans total 7,304,136 seconds (84.54 house-days), or 36.55 percent. These sums are not one wall-clock recording interval. The longest consecutive mains gap in house 5 is 2,714,841 seconds, about 31.42 days. Timestamps alone do not identify whether such gaps arose from logging, communications, intentional collection boundaries, or another cause.

3 Exact timestamp pairing is different from duration coverage

The new importer retains the union of native observations: 7,755,931 timestamps, including incomplete rows with explicit masks. Both mains legs are available at 7,056,896 timestamps; all 116 required channel streams, considered within their respective homes, coincide at 1,797,457 timestamps. The pooled exact-complete fraction of both-mains timestamps is 25.47 percent. Different cadence and timestamp phase contribute to this ratio; it must not be relabeled as 74.53 percent sensor data loss. [14]

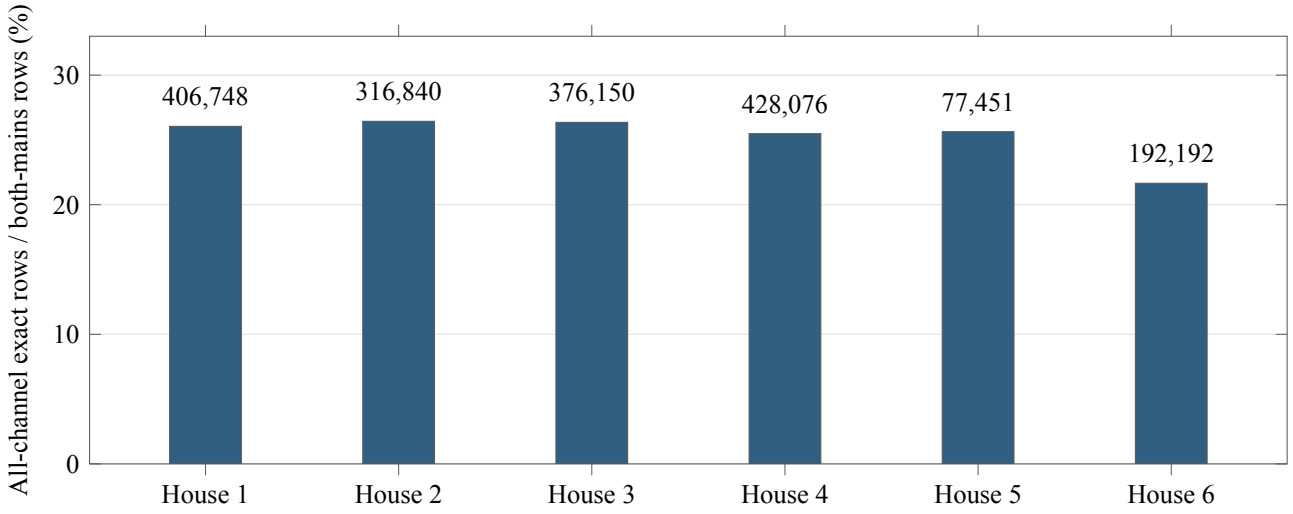


Figure 2: Exact all-channel coincidence divided by both-mains timestamp count, with complete-row counts above the bars. This is a count ratio, unlike the duration measure in Figure 1. The new importer preserves all union rows rather than dropping incomplete timestamps.

Within each house, the two mains timestamp sets are identical and every circuit has the same timestamp set as every other circuit. Thus every per-target exact pairing count equals the all-target count in this release. No missing constituent was hidden by the old any-member group rule, including the second mains leg. These observations narrow the earlier code audit: its synthetic failure cases are real conditional behaviors, not demonstrated corruption in this archive.

House 2 circuit recording begins 22,393 seconds (6 h 13 min 13 s) after mains recording and ends eight seconds earlier. This is a boundary difference, not evidence of a constant six-hour clock offset. Observed modal intervals are descriptive: house 6 mains are most often two seconds apart, and house 1/5 circuits four seconds apart, despite nominal one-/three-second descriptions.

The largest 30-to-300-second change in shared duration is 0.4535 percentage points, in house 3. The main coverage conclusion is therefore stable across these declared absolute thresholds. Cadence-relative sensitivities and per-channel exact spacing distributions remain available in the supporting data; no threshold is a certified outage boundary.

4 Numerical balance diagnostics with unchanged mains

At complete exact timestamps, define the numerical residual as MAIN_1 plus MAIN_2 minus the sum of the complete historical target groups. The primary diagnostic retains its sign and excludes no negative value. There are 2,859 negative cases among 1,797,457 eligible timestamps, or 0.1591 percent; the remaining 1,794,598 residuals are positive. All original mains readings remain intact. [14]

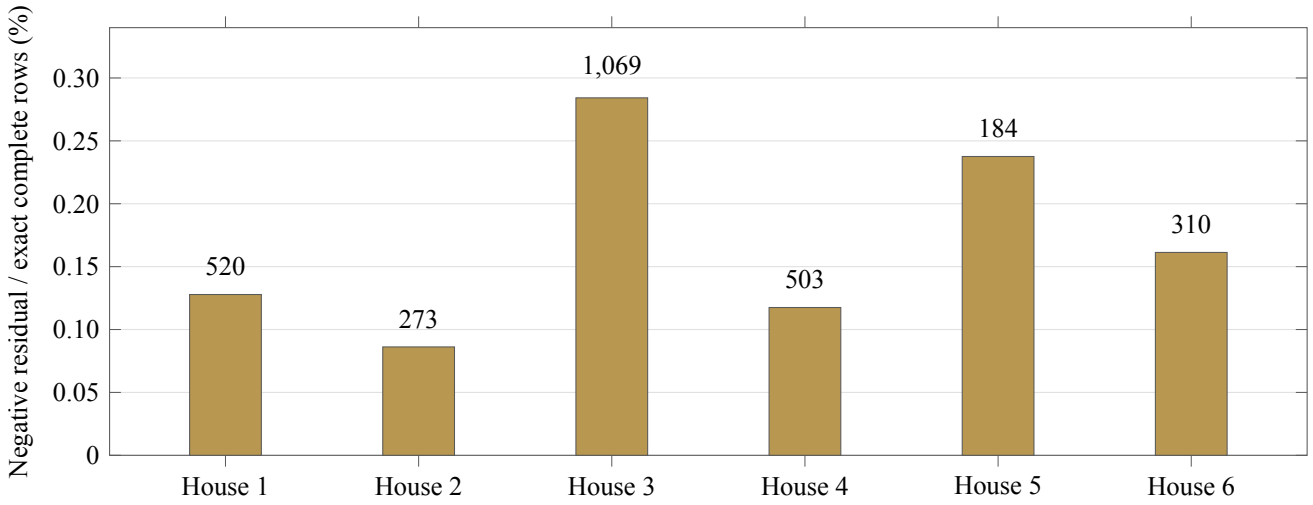


Figure 3: Frequency of negative numerical residuals in exact complete rows; labels show case counts. This is not a frequency of verified meter errors or clock faults. The current importer retains every case without replacing mains.

Table 3: Magnitudes conditional on a negative residual, in unresolved source power units

House	Complete rows	Negative rows	Median magnitude	90th percentile	Maximum magnitude
1	406,748	520	117.34	2,060.81	4,082.94
2	316,840	273	203.40	1,019.72	2,343.25
3	376,150	1,069	413.72	1,433.35	5,415.11
4	428,076	503	75.39	641.18	2,347.34
5	77,451	184	44.37	742.26	2,961.25
6	192,192	310	66.44	902.45	2,684.69

Across the negative subset, median magnitude is 125.19, the 90th percentile 1,421.24, and maximum 5,415.11 source power units. These are conditional imbalance magnitudes, not calibration errors. Supporting tables also disclose absolute and relative sensitivity cutoffs and their eligible denominators.

The original README calls low-frequency readings apparent power, whereas NILMTK conventionally labels circuit readings active power. That documentation discrepancy remains unresolved by this timestamp/value audit. Meter type, averaging windows, circuit topology, unmonitored loads, and timing can all affect the residual. No watt-hour, unknown-energy, physical synchronization, or NILM accuracy claim is inferred from these sparse exact-time sums.

5 What the REDD benchmark actually contains

REDD was an important contribution to open energy-disaggregation research. Its original system combined high-frequency whole-house waveforms with lower-rate circuit and plug measurements. The detailed hardware description reports circuit polling at approximately three seconds and whole-house waveforms at 15 kHz. Processed whole-house power is available at nominally one-second intervals. REDD therefore cannot accurately be described as having uniformly synchronized one-hertz appliance ground truth. [1, 2]

The conventional public low-frequency release contains six homes. The original 2011 paper described ten monitored homes and 119 combined days at the time of writing; those collection figures must not be interpreted as 119 days for each released home. Usable duration also depends on the selected channels and the definition of a gap. The NILMTK diagnostic study reports a house-level median mains uptime of 18 days, with a range of 4 to 19 days under its procedure. These are diagnostic results, not universal durations for every REDD export. [1, 2]

A calendar span is not the same as valid evaluation exposure. A file can begin in April and end in May while containing long outages, short dropouts, and intervals with no usable target measurement. Intersecting mains with several targets can leave less data than any channel alone. Researchers should report the calendar span, observed duration, valid shared duration, and appliance activation counts separately.

Nor is every labeled circuit an individual appliance. A kitchen-outlet or lighting circuit may combine several devices. Recovering that circuit is a legitimate task, but a circuit-level score does not demonstrate recognition of each appliance connected to it. The original REDD descriptions and experiments include such grouped loads. [1]

These limitations matter because a small set of repeated operating cycles can contain many correlated samples. Thousands of overlapping windows from one refrigerator are not thousands of independent tests of household generalization. Short records also provide little evidence about seasons, appliance replacement, changing occupancy, and rare operational states. Those are limits on the conclusions an experiment can support, even when its numerical score is computed correctly.

6 Timing and missing readings change the experiment

Different sampling cadences, unequal recording boundaries, and true clock errors are separate problems. A UTC timestamp alone does not prove that two meters integrate power over the same physical interval. The INDiC study notes that REDD house 2 appliance recording began about six hours after mains recording. That is evidence of different recording boundaries, not evidence of a universal six-hour clock offset. The official NILMTK converter also sorts REDD records because the raw data is not always ordered. [3, 6]

Joining channels by row number is invalid when timestamps differ. Nearest-neighbor joins, forward filling, interpolation, and bin averaging each impose different assumptions about when a measurement applies. Repeating a three-second appliance observation onto a one-second grid increases the number of labels; it does not create three independent one-second observations. Transition times remain uncertain at the native measurement scale.

A simple calculation shows why this matters. Consider a constructed 1 kW rectangular activation lasting 120 seconds. An otherwise exact prediction shifted by four seconds has 8,000 watt-seconds of absolute error, or 2.22 Wh. Its samplewise ON-state F1 is about 0.967 on an otherwise OFF background. For an activation lasting only ten seconds, the same shift reduces that samplewise F1 to 0.60. These are illustrative timing effects, not an estimate of REDD clock error. An algorithm can be penalized by uncertain ground truth, while a generous matching tolerance can conceal the same uncertainty.

Missing readings are independently documented. NILMTK illustrates extended mains outages and recurring circuit dropouts. Its published REDD diagnostics report a median 10 percent dropout rate when long gaps are excluded and a median 40 percent mains uptime relative to the relevant recording span. Both quantities depend on the diagnostic definitions and channels; neither should be advertised as a universal REDD missingness rate. [2]

Zero filling asserts that an unobserved appliance was off. Forward filling asserts that its previous state persisted. Linear interpolation can invent a gradual transition. Dropping rows can remove difficult intervals; treating the retained rows as contiguous can join events that were hours apart. Event pairers, recurrent models, and duration-based classifiers are especially sensitive to these changes. The direction of bias is conditional: corrupted labels can depress performance, whereas selective exclusion or smoothing can make a task easier.

Preserve observation masks and original timestamps, split sequences at excluded gaps, and document model-state handling. Report excluded observed target energy and activation counts; energy during absent-reference periods remains unknown unless estimated explicitly. Imputed values may support inference but should not silently become measured ground truth.

7 A historical importer case study

This section examines Stephen Makonin's own DataWrangle_REDD importer at commit 66ea7e0, dated 30 July 2015. Its grid, grouping, exclusion, and negative-residual repair rules are design choices of that historical software, rather than attributes of REDD's acquisition system. The code places observations on an integer-second grid, groups channels, skips rows without a mains timestamp or acceptable grouped fields, and attempts to repair negative aggregate residuals. The archived import log is a record of this preparation attempt. [4, 5]

Table 4: Historical export counts recorded in DataWrangle_REDD

House	One second grid positions	Exported rows	Retained percent	Lead and lag substitutions	Residual rebuilds
1	3,134,094	406,748	12.98	450	74
2	3,026,450	316,840	10.47	249	25
3	3,870,508	376,150	9.72	906	156
4	4,145,367	428,076	10.33	427	78
5	3,786,978	77,451	2.05	163	19
6	2,022,746	192,192	9.50	187	124

Retained percent is exported rows divided by the log's dense one-second grid length. Lead and lag substitutions combine two counters in the log; residual rebuilds correspond to its negative-noise counter. These are pipeline counts, not independently verified clock faults, sensor failure rates, continuous recording days, or model scores. [5]

The denominator is essential. Much of the grid has no exact circuit observation because the native circuit cadence is approximately three seconds; outages and nonmatching availability remove further rows. Thus the reported 87 to 98 percent shrinkage must not be relabeled as 87 to 98 percent sensor data loss. Completeness is also checked after channel grouping, so a passing row does not establish that every original constituent channel was present. [4, 5]

The log names its exports with a 3sec suffix while the inspected 2015 script uses lowf. Exact historical execution identity therefore remains unproven. The fresh raw audit now supplies separate file hashes and quality results, and

its complete-row counts match the archived export counts. Matching counts do not retrospectively establish the byte identity of the 2015 inputs or the exact program version that generated the log.

8 Repair can compromise the independence of a test

The historical DataWrangle_REDD snapshot can change measured mains using submeter information. When summed submeter measurements exceed the mains reading, the script attempts to search neighboring mains at nominal offsets of one through nine seconds and substitutes a value that makes the residual nonnegative. If that search fails, it rebuilds mains from the target sum plus an estimated residual. This describes a heuristic in the archived code, not a verified timing correction or REDD's acquisition procedure. The operations are visible in lines 129 to 163 of the pinned script. [4]

This approach may help investigate inconsistent records, but it cannot independently prove the physical cause of a negative residual. The condition can reflect timing differences, incompatible electrical quantities, meter error, channel mapping, or incomplete measurement definitions. A successful substitution is a successful heuristic repair, not an independently measured clock correction. For example, the log records an imported mains value of 172 and grouped submeter sum of 1,801 at timestamp 1303207434, followed by a replacement mains value of 1,837. This establishes an imbalance in the imported representation, without isolating its physical cause. [5]

For blind NILM evaluation, the methodological consequence is that information from the target measurements can determine the input presented to the model. Rebuilding aggregate input from the same submeters used as ground truth introduces a dependence that an ordinary deployed system does not have. Searching future mains values also uses evidence unavailable at the original timestamp. Such a track should be disclosed as retrospective, target-informed repair, with results reported separately from tests on independently measured mains.

The archive is valuable because it makes its transformations inspectable. The log records 2,858 mains substitutions or rebuilds across 1,797,457 exported rows, approximately 0.159 percent. These sparse row counts do not establish that repair explains a large performance improvement; the impact depends on event location and the energy affected. This case study does not establish that any particular REDD publication used this repository or these repair rules.

The broader issue also arises when researchers construct an aggregate by summing selected targets. A controlled aggregate can test source separation under known conditions, but it removes unmonitored loads and couples the input to its labels. Klemenjak, Makonin, and Elmenreich found that testing on denoised aggregates usually improved performance in their study, with exceptions. Their results do not imply a universal inflation factor or a simple monotonic relationship between residual magnitude and difficulty. [11]

A strong paper therefore reports two clearly identified tracks where useful: independently measured mains for deployment-oriented evidence, and constructed or repaired aggregates for controlled analysis. It preserves raw observations and publishes the locations of every input change, whether target labels influenced that change, and whether the operation requires future data.

9 Measurement compatibility and the meaning of ground truth

Low sampling rate and inaccurate measurement are not synonyms. A well-calibrated one-minute meter may provide reliable energy measurements while missing switching detail. A high-rate stream may preserve transitions while suffering calibration or timing errors. A dataset comparison must distinguish electrical accuracy, temporal resolution, synchronization, completeness, and population coverage.

Electrical quantity is particularly important. NILMTK conventionally maps REDD mains channels to apparent power and circuit channels to active power. Its metadata acknowledges inconsistent source documentation about the circuit meter. Researchers should therefore verify the convention used in their actual files rather than assert that every REDD export has a metrologically established active-versus-apparent mismatch. [3]

Active power P, measured in watts, and apparent power S, measured in volt-amperes, cannot simply be substituted for one another in an additive energy model. If mains and target measurements are incompatible, the residual between mains and the sum of targets is not a clean measure of unknown appliances. Even with compatible units, that residual can include timing error, incomplete circuit coverage, double counting, and meter error. Its sign and magnitude require diagnosis.

For this reason, an experiment should identify both split-phase mains channels, its channel aggregation rules, the measurement type of each stream, and any voltage or power-factor assumptions. A balance check should be run on valid aligned intervals and should retain negative residuals as diagnostic evidence rather than automatically forcing them to zero.

Ground truth also has a scope. A metered circuit is a reference for that circuit; it is not necessarily a reference for each device on it. An interpolated appliance trace is a reconstruction; it is not a new physical observation. A cluster assigned an appliance name after comparison with test labels demonstrates clustering agreement under that mapping rule. It does not by itself demonstrate autonomous appliance identification.

The recommendation is to choose complementary reference data by the property being tested. Measurement quality should support energy-error claims. Timing and native sampling should support event claims. Independent homes and longer records should support generalization claims. All three are needed when a system is presented as ready for practical deployment.

10 Complementary datasets and evaluation roles

The following comparison uses associated journal descriptions, supplemented by the author-maintained R1Hz technical documentation. It identifies useful evidence for different evaluation purposes, without a universal accuracy ranking. The author's involvement in AMPds2, RAE, and R1Hz is disclosed above; their own coverage and recovery limits remain part of the comparison. Each actual release requires a version and quality audit. [7, 8, 9, 10, 12, 13]

Table 5: Dataset selection by the property requiring validation

Dataset	Contribution to validation	Limit that must remain visible
AMPds2 [7]	One Canadian home over 730 days at one-minute intervals. Industrial DENT meters have specified 1 percent accuracy and less than 0.5 percent typical error. Useful for longitudinal energy and multistate-load tests.	One home limits population inference. One-minute readings cannot resolve fast events. Timestamp corrections and filled readings must be accounted for.
R1Hz [13]	A 757-day calendar span at one Burnaby, Canada dwelling, with five aligned one-hertz electrical streams, 19 logical channels, interval-energy products, utility comparisons, and retained acquisition records. Useful for longitudinal energy and temporal-resolution tests.	One dwelling and mixed circuits limit generalization. Retain all quality markers, including s and +; exclude imputed references from primary ground-truth scoring or assess them separately. Calendar coverage is not uninterrupted observation.

Dataset	Contribution to validation	Limit that must remain visible
UK-DALE [8]	Five homes. Houses 1, 2 and 5 add one-hertz active-power mains and 16 kHz waveforms. Selected mains systems were individually calibrated; a resistive-load test reported less than 2 percent relative error.	The reference meter also has uncertainty. Appliance observations are approximately six seconds apart, and coverage is incomplete. Mains calibration does not validate every target meter.
REFIT [9]	Twenty homes over approximately two years, with aggregate and nine selected appliance streams per home at eight-second intervals. Useful for household diversity and unknown-load robustness.	A breadth test, not a precision reference. Its paper documents current-only measurement, asynchronous readings, outages, and imputation.
RAE [10]	The original two-home release includes up to 24 circuits at one hertz for 72 and 59 days, plus independent utility observations at roughly eight to fifteen seconds.	The one-hertz mains column can be calculated from submeters or separately measured. Metadata must identify which; constructed and measured aggregates need separate evaluation tracks.
IDEAL [12]	A larger optional robustness test. The study includes 255 homes; an enhanced subset of 39 adds selected appliance and active-power monitoring.	Appliance ground truth is not available for all 255 homes. Missingness and the difference between ordinary and enhanced instrumentation require subset-specific auditing.

AMPds2 and R1Hz are useful complementary choices for longitudinal energy evaluation. AMPds2 cannot establish second-level latency; R1Hz adds one-second measurements over a longer calendar span but requires explicit handling of recovered intervals. Neither single-home dataset establishes population generalization. UK-DALE supports selected calibrated-mains comparisons, with appliance timing still requiring attention. REFIT tests more homes and unmonitored loads, while introducing its own data-quality constraints. RAE requires a distinction between measured and calculated aggregate channels. These roles are complementary.

11 A minimum standard for credible NILM experiments

We recommend the following requirements for papers making broad claims about NILM accuracy, generalization, or real-time use. These are proposed evaluation requirements, not claims that an existing dataset satisfies them automatically.

Identify the physical test signal. Publish the dataset release, raw-file checksums, home and channel identifiers, electrical quantities and units, selected dates, and whether the aggregate is independently measured, reconstructed, or target-informed. Record the software revision and complete configuration.

Publish a data-quality audit. Report native sample-interval distributions, duplicates, ordering faults, outages, short-dropout rates, and valid shared exposure. Supply masks for observed, imputed, excluded, and repaired values. Give valid duration, activation counts, and target energy by home and appliance; document every exclusion.

Freeze the preprocessing and split rules. Estimate any alignment correction or calibration on designated training or calibration intervals, then freeze it before testing. State interpolation limits and bin boundaries. Split sequences at gaps and prevent overlapping windows or complete activations from crossing train, validation, and test boundaries.

Test independent homes and datasets. Separate same-home chronological testing from held-out-home testing. Keep test labels out of model selection and input repair. Add a dataset chosen for measurement quality and another chosen for household diversity or realistic unmonitored loads. Report failures as well as pooled averages.

Match resolution without inventing observations. Report native-resolution performance and a common-resolution comparison where appropriate. Use a documented downsampling rule that respects measured intervals. Do not represent upsampled ground truth as new temporal evidence or claim an event timestamp more precise than the reference supports.

Evaluate on shared valid intervals. Apply the same predeclared observation masks to every algorithm. Score imputed or repaired regions separately. Pair power MAE with energy error, activation precision and recall, and complete per-appliance results. Include simple zero-output, constant-output, and feature-matched baselines where appropriate.

Measure when predictions become available. Log the first causal prediction and each later revision. Include observation, settling, window, and cycle-completion delays in reported latency. A model that identifies an activation after switch-off should be assessed for that completed-cycle task, not credited with immediate switch-on identification.

Make the statistical unit explicit. Estimate household-level uncertainty for new-household claims. Use nested temporal or activation blocks for within-household repeatability, preserving dependence. Do not treat densely sampled points or overlapping windows as independent replicates. Report sensitivity to gap thresholds, alignment tolerances, preprocessing, and important parameter choices.

12 How reviewers should interpret REDD results

A reproducible REDD result can support a limited algorithmic claim: the method performs as stated on the disclosed channels, time periods, electrical quantities, and quality masks. A study of a controlled constructed aggregate can support a source-separation claim under those construction assumptions. Neither result alone establishes generalization across acquisition systems or the wider household population.

A deployment-oriented claim requires more. The result should survive a second measurement system, a broader set of homes or operating conditions, explicit gap handling, and an evaluation of output availability. High accuracy on one or two familiar REDD homes should therefore prompt scrutiny of data preparation and task definition before it is credited as evidence of a broadly superior NILM system.

Reviewers should request clarification when a paper describes all REDD streams as one hertz, omits the usable shared duration, gives no policy for missing readings, confuses circuit labels with appliances, or reports noise ratios without checking compatible measurements. Claims of real-time use also require the observation delay, while claims of autonomous appliance recognition require a documented label-mapping procedure.

The same standards must apply to alternatives. A longer or newer dataset can still contain imputed values, asynchronous meters, uncertain units, and incomplete appliance coverage. It is the combination of transparent measurements, reproducible processing, appropriate claims, and independent tests that makes the evidence credible.

REDD continues to support historical reproduction and targeted algorithmic evaluation. When claims extend to practical operation across households or measurement systems, researchers should show data integrity and test the claimed advantage under complementary conditions. Transparent processing and evidence proportionate to the claim allow subsequent work to build on REDD's contribution.

13 Historical log context: a separate processing record

The archived 2015 log describes Stephen Makonin's exploratory importer and its target-informed changes to mains. It is not REDD's original acquisition pipeline and does not describe the current Python 3 import. The historical exclusion logic and repair behavior remain useful methodological examples, with the execution-version qualification stated above. [4, 5]

The archive records 1,381 lead substitutions, 1,001 lag substitutions, and 476 fallback rebuilds: 2,858 changes across 1,797,457 exports, or 0.1590 percent. The 476 detailed rebuilds have median MAIN increase 301, 90th percentile 1,772.5, and maximum 5,116 in archived integer power units. Those changes were made by the importer; they are not measured acquisition errors.

The fresh raw audit instead finds 2,859 negative residuals using retained decimal readings and no repairs. These counts are not interchangeable: they describe different arithmetic and processing states, and the exact historical execution is unknown. No particular NILM score effect follows from either count.

Every house's fresh all-channel exact-complete count equals its historical exported count: 406,748; 316,840; 376,150; 428,076; 77,451; and 192,192. No malformed/duplicate data, partial-group concealment, missing second mains leg, or first/last timestamp-extrema failure was observed in this canonical release. These specific partial-group, duplicate, and endpoint failure modes have no observed trigger in this copy. This does not certify all historical repair logic or reverse its logged changes to mains.

Original historical-log tables, all 476 paired rebuild records, and editable historical plots remain in the supporting sources for reproduction. The fresh audit has its own provenance, protocol, masks, checks, and statistics. Large raw files, observation exports, row masks, and the local audit database are not redistributed with this note.

The defensible conclusion is specific: this archive has clean numeric record structure and reproducible timestamp ordering, alongside substantial recording gaps, cadence variation, and limited household/temporal coverage. Those are reasons to state a benchmark's scope and seek complementary validation, not to invalidate REDD or its research contribution.

References

[1] J. Zico Kolter and M. J. Johnson. REDD: A Public Data Set for Energy Disaggregation Research. Workshop on Data Mining Applications in Sustainability, SustKDD, 2011. Original dataset paper; no DOI identified. Author-hosted paper.

[2] N. Batra, J. Kelly, O. Parson, H. Dutta, W. Knottenbelt, A. Rogers, A. Singh, and M. Srivastava. NILMTK: An Open Source Toolkit for Non-intrusive Load Monitoring. Proceedings of the 5th International Conference on Future Energy Systems (e-Energy 2014), 2014, pp. 265-276. REDD diagnostics in Section 4 and Table 2. DOI 10.1145/2602044.2602051.

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[4] S. Makonin. DataWrangle_REDD Import_REDD_dataset.py. Source snapshot 66ea7e01e499be2bb6b88dbaf8f3cee3445cbe0d, 30 July 2015. Repository code is copyright 2013. Grid and exclusion logic at lines 72-123; repair logic at lines 129-163. Pinned source code.

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